-Latent semantic analysis operates using singular value decomposition. The purpose of SVD is to factorize a real or a complex matrix. Through SVD a matrix A is decomposed in 3 matrix, U Σ, and V.

-SVD properties are : $U^T U = V^T V = I$

-Using SVD there are several ways to reduce matrixes, thin SVD's transforms Σ to a square matrix and removes the corresponding columns of U

-Compact SVD reduces Σ by removing zero-singular values and corresponding vectors in U and V

- Latent Semantic Indexing is the process of term-document matrix to expose statistical structure by converting a high dimensional space to lower dimensional space.

-It can be used to reduce the number of features and documents.

-Semantic: represent meaning as a function of similarity to other entities.

-To reduce the dimension using SVD : step 1 : Construct the approximated matrix A_k from the original term-document matrix A using SVD

$$\mathbf{A} \approx \mathbf{A}_k = \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^T$$

-Step 2 : Multiply the transpose of U_k to obtain k (<<m) by n term-document matrix

$$\mathbf{U}_k^T \mathbf{A}_k = \mathbf{U}_k^T \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^T = \mathbf{I} \mathbf{\Sigma}_k \mathbf{V}_k^T = \mathbf{\Sigma}_k \mathbf{V}_k^T$$

-Step 3: Apply data mining algorithms

-Stochastic neighbor embedding emphasizes that it is more important to get local distances right than non-local ones.

- We need to use different radii in different parts of the space so that we keep the effective number of neighbors about constant

- Decide what entropy you want and then find the radius that produces that entropy

-the performance of SNE is fairly robus to change in perplexity (varies between 5 and 50)

- Kullback-Leibler divergence is a non-symmetric measure of the difference between two probability distribution P and Q

$$Cost = \sum_i K\, L(P_i || Q_i) = \sum_i \sum_j p_{j|i} \log \frac{p_{j|i}}{q_{j|i}}$$

Pj|t != pt|j not symmetric, pij = pj|I + pi|j /2n

-but it can suffer from crowding problem : The area accommodating moderately distant data points is not large enough compared with the area accommodating nearby data points

- the solution is to use a probability distribution that has much heavier tails than a Gaussian to convert distances into probabilities in the low-dimensional map